

3.2.5

AI25BTECH11004-B.JASWANTH

Question:

Draw a triangle ABC in which BC = 6 cm, CA = 5 cm and AB = 4 cm.

Solution:

→ Let

$$a = \|\mathbf{C} - \mathbf{B}\| = 6\text{cm} \quad (0.1)$$

$$b = \|\mathbf{A} - \mathbf{C}\| = 5\text{cm} \quad (0.2)$$

$$c = \|\mathbf{B} - \mathbf{A}\| = 4\text{cm} \quad (0.3)$$

→ By using cosine law in $\triangle ABC$, we get

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} \quad (0.4)$$

$$\Rightarrow \cos B = \frac{6^2 + 4^2 - 5^2}{2 \times 6 \times 4} \quad (0.5)$$

$$\Rightarrow \cos B = \frac{9}{16} \quad (0.6)$$

$$\Rightarrow \angle B = \cos^{-1}\left(\frac{9}{16}\right) \approx 55^\circ \quad (0.7)$$

→ The coordinates of $\triangle ABC$ can then be expressed as

$$\mathbf{A} = c \begin{pmatrix} \cos B \\ \sin B \end{pmatrix} \quad (0.8)$$

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.9)$$

$$\mathbf{C} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \quad (0.10)$$

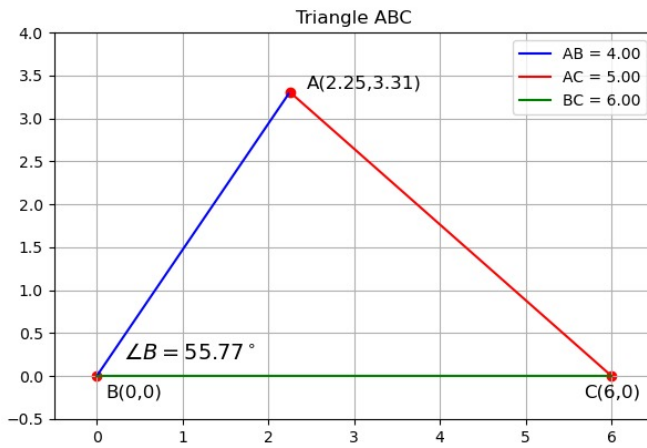


Fig. 0: Plot of $\triangle ABC$